IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

# RADIO FREQUENCY SUBSYSTEMS

Tuesday, 5 May 2026, 10:00 AM

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer Question ONE and any TWO out of the remaining three questions.**

*Question ONE is COMPULSARY and carries a weighting of 50%. The remaining questions carry a weighting of 25% each.*

*This is an OPEN BOOK examination.*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : W.T. Pike

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## Special instructions for invigilators

*This is an OPEN BOOK examination.*

## Special instructions for students

*There are 4 (four) questions in total.*

*Question 1 is COMPULSARY and carries a weighting of 50%.*

*Answer 2 (two) out of the remaining 3(three) questions, each carrying a weighting of 25%.*

*All variables and acronyms have their usual meaning.*

# The Questions

1.

a) Consider the satellite communications link illustrated in Figure 1.1. The satellite is in a circular geosynchronous orbit in the plane of the Earth's equator and has an altitude of 35,786 km above sea level. Both the uplink and downlink earth stations are both located on the equator, at sea level. For simplicity, ignore the curvature of the earth.

i) Write down the well-known Friis range equation for a point-to-point communications link.

[3]

ii) From 1(a)(i), deduce the expression for the free space path loss term shown in Figure 1.1 and use this to calculate the uplink and downlink ranges.

[5]

iii) What is the separation distance between the uplink and downlink earth stations? *Hint: Use simple trigonometry.*

[4]

iv) If the uplink and downlink earth station antennas are now pointed directly towards each other and the downlink earth station is retuned to operate at 6 GHz, calculate its LNA gain to achieve the originally specified output signal power.

[4]

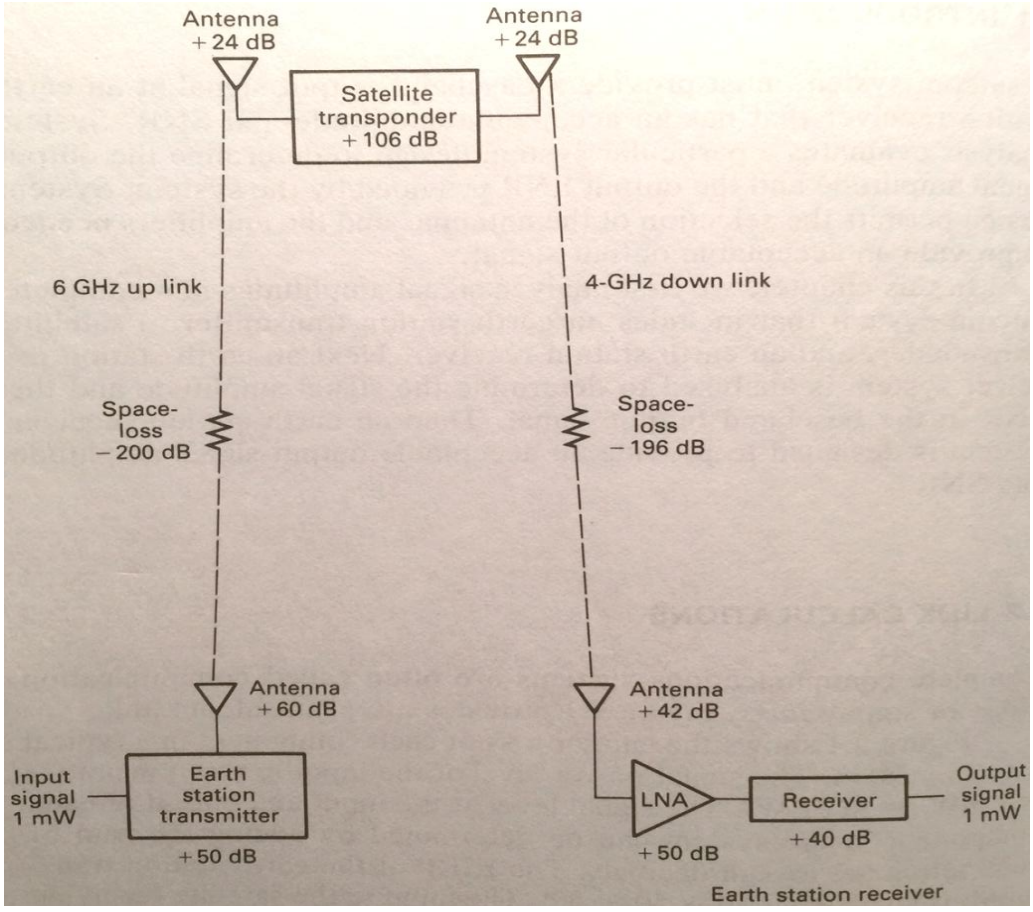



Figure 1.1 (taken from R.L. Douglas, *Satellite Communications Technology*).

*Question 1 continues onto the next page…*

b) Consider a signal generator connected to a load impedance, via a transmission line having a power loss $L_L$, all being matched to the same reference impedance. Assuming the signal generator, transmission line and load impedance are all in thermodynamic equilibrium and at the physical ambient temperature $T_o$. The input signal power $S_i$ and input noise power $N_i$ to the transmission line can be combined to give an input signal-to-noise power ratio $SNR = S_i/N_i$.

i) Give the mathematical relationship between equivalent noise temperature and noise factor.

[3]

ii) Give an expression for the output noise power.

[3]

iii) If the transmission line has a loss of 3 dB, what is its equivalent noise temperature in relation to $T_o$?

[4]

iv) Determine the output signal power $S_o$ and output noise power $N_o$ from the transmission line and give the output signal-to-noise power ratio $SNR = S_o/N_o$. What is the mathematical relationship between the output and input SNR?

[4]

c) Consider the downlink earth station in Figure 1.1.

i) Write down the well-known Friss formula for the noise factor of cascaded stages.

[3]

ii) Using the relationship in 1(b)(i) with that in 1(c)(i), derive the equivalent noise temperature for two cascaded stages.

[2]

iii) If the equivalent noise temperatures for the antenna, LNA and receiver are 38 K, 120 K and 4,000 K, respectively, calculate the system equivalent noise temperature.

[4]

iv) With Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K and a receiver bandwidth of 27 MHz, calculate the noise power at the output of the receiver.

[4]

v) With the satellite having an EIRP of +34 dBW, calculate the signal power at the output of the receiver.

[4]

vi) Calculate the SNR at the output of the receiver.

[3]

2.

a) Draw the block diagram for a single-conversion superhet receiver and, stating its full name, explain the origin of this type of architecture.

[3]

b) For the receiver architecture in 2(a):

i) With the help of a spectral illustration, briefly explain why an image rejection filter is needed and where this should be located within the architecture. Also, briefly explain why this filter represents a problem and list 3 possible solutions that can be adopted to alleviate or avoid this problem.

[4]

ii) For selecting bands with higher levels of adjacent-band rejection, what simple method is usually adopted?

[2]

c) The receiver has an input power level of -50 dBm. With the simplest architecture design, for the type in 2(a) up to the demodulator, it can be assumed that there is only one filter and one amplifier at any given frequency. All amplifiers have a gain of 30 dB, $IP_3$ of 40 dBm and a noise figure of 1 dB; all filters have a pass band insertion loss of 2 dB and the mixer has a conversion loss of 6 dB and $IP_3$ of 30 dBm. Ignoring noise contributions from the local oscillator, calculate the following at the output of each sub-system block (where appropriate) and state the main equations used:

i) $C$

[2]

ii) $IP_3$ (clearly state any assumptions)

[3]

iii) $IMD_3$

[2]

iv) $I_3$

[3]

v) Overall $F_{RX}$

[3]

vi) Carrier-to-noise ratio at the output of a 30 kHz bandwidth receiver having its overall system temperature equal to the ambient temperature of 290 K.

[3]

3. a) (i) For a lossless linear two-port network, state the simple equation that show the relationship between insertion loss and return loss.

[3]

(ii) Using the equation stated in 3(a)(i), calculate the return loss when insertion loss is -1 dB.

[4]

b) Draw a 3-channel frequency de-multiplexer and a 3-channel frequency multiplexer, employing circulators, band-pass filters and dummy loads.

[5]

c) Employing the topologies given in 3(b), draw a unilateral 3-channel amplifier. Assume that for each frequency band the associated filters and amplifiers have the same -1 dB band edge frequencies.

[4]

d) Employing the topology from in 3(c), use simple power loss budget analysis to calculate the overall gain, given that the insertion loss of each circulator is - 0.1 dB, the mid-band insertion loss of the filters is 0 dB and the mid-band gain of the amplifiers is 10 dB. State any simplifying assumptions.

(i) Calculate the gain at mid-band, assuming that the transition frequency bandwidth is insignificant.

[3]

(ii) Calculate the gain at the -1 dB frequency band edge, assuming that the ideal upper -1 dB band edge frequency for one channel is the same as the ideal lower -1 dB band edge frequency for the adjacent channel.

[4]

(iii) Using the results from 3(d)(i) and (ii), comment on the resulting frequency response over the entire bandwidth of the 3 channels.

[2]

4.

a) With the use of simple illustrations for the attenuation against frequency curves, describe the differences between Butterworth, Chebyshev and Elliptical-function filters. Also, comment on the group delay characteristics for these filters.

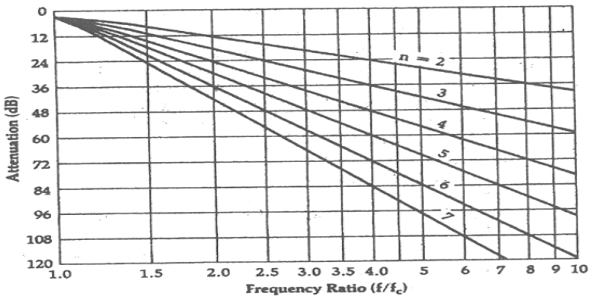
[5]

b) Given prototype low-pass filter attenuation curves and tables for the corresponding normalised element values (see attached sheets), design an L-C lumped-element band-pass filter that meets the following specifications:

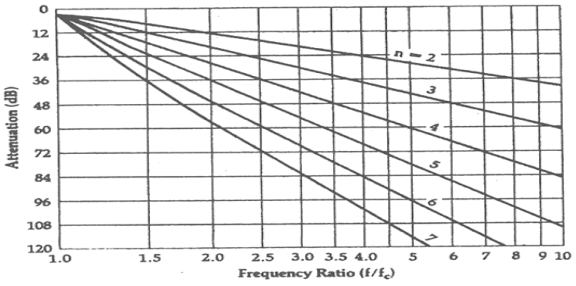
| | |
|---|---|
| Centre Frequency, *fo* | 500 MHz |
| 3 dB Bandwidth, *B* | 50 MHz |
| Attenuation Bandwidth | 100 MHz |
| Pass-Band Ripple (Peak-to-Peak) | 0.1 dB |
| Stop-Band Attenuation | 45 dB |
| Input Impedance, $R_{IN}$ | 100 Ω |
| Output Impedance, $R_{OUT}$ | 50 Ω |

[20]

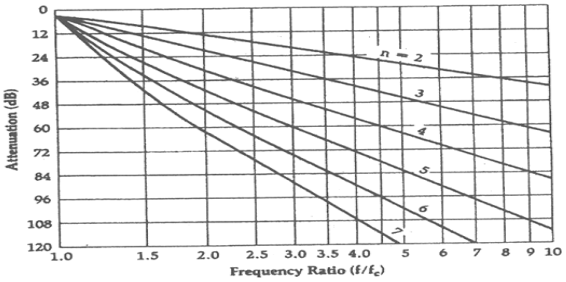
# Standard Filter Curves and Tables



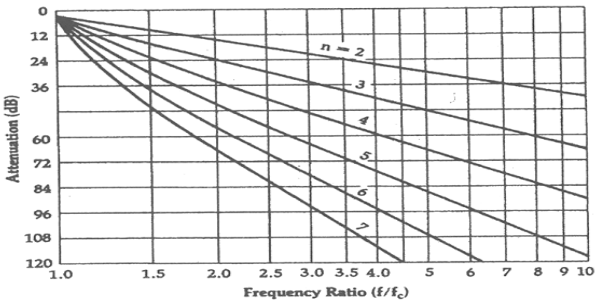

Attenuation characteristics for Butterworth filters.



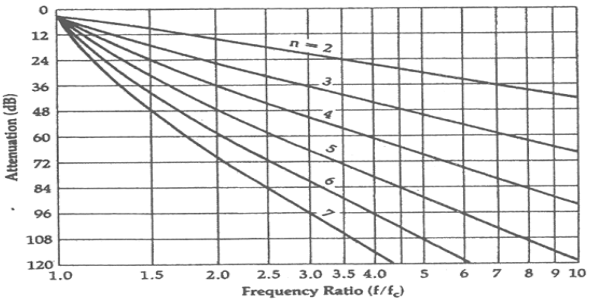

Attenuation characteristics for a Chebyshev filter with 0.01-dB ripple.




Attenuation characteristics for a Chebyshev filter with 0.1-dB ripple.

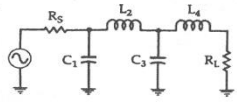



Attenuation characteristics for a Chebyshev filter with 0.5-dB ripple.



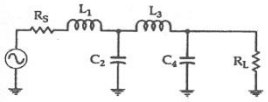

Attenuation characteristics for a Chebyshev filter with 1-dB ripple.

**Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple**

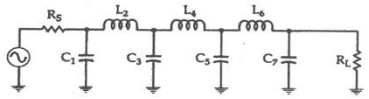



| $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ |
|---|---|---|---|---|
| 3.000 | 0.572 | 3.132 | | |
| 4.000 | 0.365 | 4.600 | | |
| 8.000 | 0.157 | 9.658 | | |
| ∞ | 1.213 | 1.109 | | |
| 1.000 | 2.216 | 1.088 | 2.216 | |
| 0.500 | 4.431 | 0.817 | 2.216 | |
| 0.333 | 6.647 | 0.726 | 2.216 | |
| 0.250 | 8.862 | 0.680 | 2.216 | |
| 0.125 | 17.725 | 0.612 | 2.216 | |
| ∞ | 1.652 | 1.460 | 1.108 | |
| 3.000 | 0.653 | 4.411 | 0.814 | 2.535 |
| 4.000 | 0.452 | 7.083 | 0.612 | 2.848 |
| 8.000 | 0.209 | 17.164 | 0.428 | 3.281 |
| ∞ | 1.350 | 2.010 | 1.488 | 1.106 |

| $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ |
|---|---|---|---|---|

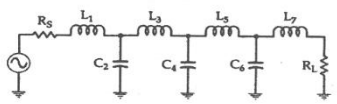



**Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple**

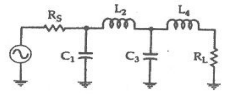



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ | $C_5$ | $L_6$ | $C_7$ |
|---|---|---|---|---|---|---|---|---|
| 5 | 1.000 | 2.207 | 1.128 | 3.103 | 1.128 | 2.207 | | |
| | 0.500 | 4.414 | 0.565 | 4.653 | 1.128 | 2.207 | | |
| | 0.333 | 6.622 | 0.376 | 6.205 | 1.128 | 2.207 | | |
| | 0.250 | 8.829 | 0.282 | 7.756 | 1.128 | 2.207 | | |
| | 0.125 | 17.657 | 0.141 | 13.961 | 1.128 | 2.207 | | |
| | ∞ | 1.721 | 1.645 | 2.061 | 1.493 | 1.103 | | |
| 6 | 3.000 | 0.679 | 3.873 | 0.771 | 4.711 | 0.969 | 2.406 | |
| | 4.000 | 0.481 | 5.644 | 0.476 | 7.351 | 0.849 | 2.582 | |
| | 8.000 | 0.227 | 12.310 | 0.198 | 16.740 | 0.726 | 2.800 | |
| | ∞ | 1.378 | 2.097 | 1.690 | 2.074 | 1.494 | 1.102 | |
| 7 | 1.000 | 2.204 | 1.131 | 3.147 | 1.194 | 3.147 | 1.131 | 2.204 |
| | 0.500 | 4.408 | 0.566 | 6.293 | 0.895 | 3.147 | 1.131 | 2.204 |
| | 0.333 | 6.612 | 0.377 | 9.441 | 0.796 | 3.147 | 1.131 | 2.204 |
| | 0.250 | 8.815 | 0.283 | 12.588 | 0.747 | 3.147 | 1.131 | 2.204 |
| | 0.125 | 17.631 | 0.141 | 25.175 | 0.671 | 3.147 | 1.131 | 2.204 |
| | ∞ | 1.741 | 1.677 | 2.155 | 1.703 | 2.079 | 1.494 | 1.102 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ | $L_5$ | $C_6$ | $L_7$ |
|---|---|---|---|---|---|---|---|---|

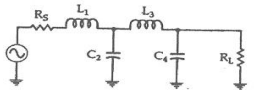


## Butterworth Low-Pass Prototype Element Values

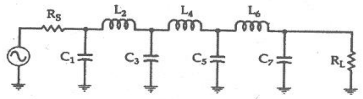



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ |
|---|---|---|---|---|---|
| 2 | 1.111 | 1.035 | 1.835 | | |
| | 1.250 | 0.849 | 2.121 | | |
| | 1.429 | 0.697 | 2.439 | | |
| | 1.667 | 0.566 | 2.828 | | |
| | 2.000 | 0.448 | 3.346 | | |
| | 2.500 | 0.342 | 4.095 | | |
| | 3.333 | 0.245 | 5.313 | | |
| | 5.000 | 0.156 | 7.707 | | |
| | 10.000 | 0.074 | 14.814 | | |
| | ∞ | 1.414 | 0.707 | | |
| 3 | 0.900 | 0.808 | 1.633 | 1.599 | |
| | 0.800 | 0.844 | 1.384 | 1.926 | |
| | 0.700 | 0.915 | 1.165 | 2.277 | |
| | 0.600 | 1.023 | 0.965 | 2.702 | |
| | 0.500 | 1.181 | 0.779 | 3.261 | |
| | 0.400 | 1.425 | 0.604 | 4.064 | |
| | 0.300 | 1.838 | 0.440 | 5.363 | |
| | 0.200 | 2.669 | 0.284 | 7.910 | |
| | 0.100 | 5.167 | 0.138 | 15.455 | |
| | ∞ | 1.500 | 1.333 | 0.500 | |
| 4 | 1.111 | 0.466 | 1.592 | 1.744 | 1.469 |
| | 1.250 | 0.388 | 1.695 | 1.511 | 1.811 |
| | 1.429 | 0.325 | 1.862 | 1.291 | 2.175 |
| | 1.667 | 0.269 | 2.103 | 1.082 | 2.613 |
| | 2.000 | 0.218 | 2.452 | 0.883 | 3.187 |
| | 2.500 | 0.169 | 2.986 | 0.691 | 4.009 |
| | 3.333 | 0.124 | 3.883 | 0.507 | 5.338 |
| | 5.000 | 0.080 | 5.684 | 0.331 | 7.940 |
| | 10.000 | 0.039 | 11.094 | 0.162 | 15.642 |
| | ∞ | 1.531 | 1.577 | 1.082 | 0.383 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ |
|---|---|---|---|---|---|

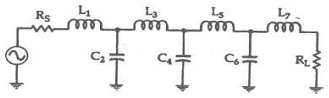



## Butterworth Low-Pass Prototype Element Values

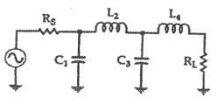



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ | $C_5$ | $L_6$ | $C_7$ |
|---|---|---|---|---|---|---|---|---|
| 5 | 0.900 | 0.442 | 1.027 | 1.910 | 1.756 | 1.389 | | |
| | 0.800 | 0.470 | 0.866 | 2.061 | 1.544 | 1.738 | | |
| | 0.700 | 0.517 | 0.731 | 2.285 | 1.333 | 2.108 | | |
| | 0.600 | 0.586 | 0.609 | 2.600 | 1.126 | 2.552 | | |
| | 0.500 | 0.686 | 0.496 | 3.051 | 0.924 | 3.133 | | |
| | 0.400 | 0.838 | 0.388 | 3.736 | 0.727 | 3.965 | | |
| | 0.300 | 1.094 | 0.285 | 4.884 | 0.537 | 5.307 | | |
| | 0.200 | 1.608 | 0.186 | 7.185 | 0.352 | 7.935 | | |
| | 0.100 | 3.512 | 0.091 | 14.095 | 0.173 | 15.710 | | |
| | ∞ | 1.545 | 1.694 | 1.382 | 0.894 | 0.309 | | |
| 6 | 1.111 | 0.289 | 1.040 | 1.322 | 2.054 | 1.744 | 1.335 | |
| | 1.250 | 0.245 | 1.116 | 1.126 | 2.239 | 1.550 | 1.688 | |
| | 1.429 | 0.207 | 1.236 | 0.957 | 2.499 | 1.346 | 2.062 | |
| | 1.667 | 0.173 | 1.407 | 0.801 | 2.858 | 1.143 | 2.509 | |
| | 2.000 | 0.141 | 1.653 | 0.654 | 3.369 | 0.942 | 3.094 | |
| | 2.500 | 0.111 | 2.028 | 0.514 | 4.141 | 0.745 | 3.931 | |
| | 3.333 | 0.082 | 2.656 | 0.379 | 5.433 | 0.552 | 5.280 | |
| | 5.000 | 0.054 | 3.917 | 0.248 | 8.020 | 0.363 | 7.922 | |
| | 10.000 | 0.026 | 7.705 | 0.122 | 15.786 | 0.179 | 15.738 | |
| | ∞ | 1.553 | 1.759 | 1.553 | 1.202 | 0.758 | 0.259 | |
| 7 | 0.900 | 0.299 | 0.711 | 1.404 | 1.489 | 2.125 | 1.727 | 1.296 |
| | 0.800 | 0.322 | 0.606 | 1.517 | 1.278 | 2.334 | 1.546 | 1.652 |
| | 0.700 | 0.357 | 0.515 | 1.688 | 1.091 | 2.618 | 1.350 | 2.028 |
| | 0.600 | 0.408 | 0.432 | 1.928 | 0.917 | 3.005 | 1.150 | 2.477 |
| | 0.500 | 0.480 | 0.354 | 2.273 | 0.751 | 3.553 | 0.951 | 3.064 |
| | 0.400 | 0.590 | 0.278 | 2.795 | 0.592 | 4.380 | 0.754 | 3.904 |
| | 0.300 | 0.775 | 0.206 | 3.671 | 0.437 | 5.761 | 0.560 | 5.258 |
| | 0.200 | 1.145 | 0.135 | 5.427 | 0.287 | 8.526 | 0.369 | 7.908 |
| | 0.100 | 2.257 | 0.067 | 10.700 | 0.142 | 16.822 | 0.182 | 15.748 |
| | ∞ | 1.558 | 1.799 | 1.659 | 1.397 | 1.055 | 0.656 | 0.223 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ | $L_5$ | $C_6$ | $L_7$ |
|---|---|---|---|---|---|---|---|---|

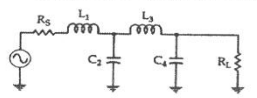



## Chebyshev Low-Pass Element Values for 0.01-dB Ripple

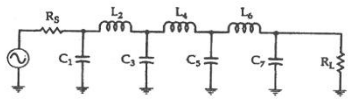



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ |
|---|---|---|---|---|---|
| 2 | 1.101 | 1.347 | 1.483 | | |
| | 1.111 | 1.247 | 1.595 | | |
| | 1.250 | 0.943 | 1.997 | | |
| | 1.429 | 0.759 | 2.344 | | |
| | 1.667 | 0.609 | 2.750 | | |
| | 2.000 | 0.479 | 3.277 | | |
| | 2.500 | 0.363 | 4.033 | | |
| | 3.333 | 0.259 | 5.255 | | |
| | 5.000 | 0.164 | 7.650 | | |
| | 10.000 | 0.078 | 14.749 | | |
| | ∞ | 1.412 | 0.742 | | |
| 3 | 1.000 | 1.181 | 1.821 | 1.181 | |
| | 0.900 | 1.092 | 1.660 | 1.480 | |
| | 0.800 | 1.097 | 1.443 | 1.806 | |
| | 0.700 | 1.160 | 1.228 | 2.165 | |
| | 0.600 | 1.274 | 1.024 | 2.598 | |
| | 0.500 | 1.452 | 0.829 | 3.164 | |
| | 0.400 | 1.734 | 0.645 | 3.974 | |
| | 0.300 | 2.216 | 0.470 | 5.280 | |
| | 0.200 | 3.193 | 0.305 | 7.834 | |
| | 0.100 | 6.141 | 0.148 | 15.390 | |
| | ∞ | 1.501 | 1.433 | 0.591 | |
| 4 | 1.100 | 0.950 | 1.938 | 1.761 | 1.046 |
| | 1.111 | 0.854 | 1.946 | 1.744 | 1.165 |
| | 1.250 | 0.618 | 2.075 | 1.542 | 1.617 |
| | 1.429 | 0.495 | 2.279 | 1.334 | 2.008 |
| | 1.667 | 0.398 | 2.571 | 1.128 | 2.461 |
| | 2.000 | 0.316 | 2.994 | 0.926 | 3.045 |
| | 2.500 | 0.242 | 3.641 | 0.729 | 3.875 |
| | 3.333 | 0.174 | 4.727 | 0.538 | 5.209 |
| | 5.000 | 0.112 | 6.910 | 0.352 | 7.813 |
| | 10.000 | 0.054 | 13.469 | 0.173 | 15.510 |
| | ∞ | 1.529 | 1.694 | 1.312 | 0.523 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ |
|---|---|---|---|---|---|

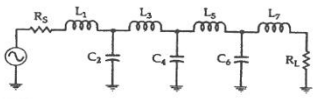



## Chebyshev Low-Pass Element Values for 0.01-dB Ripple

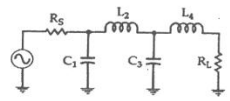



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ | $C_5$ | $L_6$ | $C_7$ |
|---|---|---|---|---|---|---|---|---|
| 5 | 1.000 | 0.977 | 1.685 | 2.037 | 1.685 | 0.977 | | |
| | 0.900 | 0.880 | 1.456 | 2.174 | 1.641 | 1.274 | | |
| | 0.800 | 0.877 | 1.235 | 2.379 | 1.499 | 1.607 | | |
| | 0.700 | 0.926 | 1.040 | 2.658 | 1.323 | 1.977 | | |
| | 0.600 | 1.019 | 0.863 | 3.041 | 1.135 | 2.424 | | |
| | 0.500 | 1.166 | 0.699 | 3.584 | 0.942 | 3.009 | | |
| | 0.400 | 1.398 | 0.544 | 4.403 | 0.749 | 3.845 | | |
| | 0.300 | 1.797 | 0.398 | 5.772 | 0.557 | 5.193 | | |
| | 0.200 | 2.604 | 0.259 | 8.514 | 0.368 | 7.826 | | |
| | 0.100 | 5.041 | 0.127 | 16.741 | 0.182 | 15.613 | | |
| | ∞ | 1.547 | 1.795 | 1.645 | 1.237 | 0.488 | | |
| 6 | 1.101 | 0.851 | 1.796 | 1.841 | 2.027 | 1.631 | 0.937 | |
| | 1.111 | 0.760 | 1.782 | 1.775 | 2.094 | 1.638 | 1.053 | |
| | 1.250 | 0.545 | 1.864 | 1.489 | 2.403 | 1.507 | 1.504 | |
| | 1.429 | 0.436 | 2.038 | 1.266 | 2.735 | 1.332 | 1.899 | |
| | 1.667 | 0.351 | 2.298 | 1.061 | 3.167 | 1.145 | 2.357 | |
| | 2.000 | 0.279 | 2.678 | 0.867 | 3.768 | 0.954 | 2.948 | |
| | 2.500 | 0.214 | 3.261 | 0.682 | 4.667 | 0.761 | 3.790 | |
| | 3.333 | 0.155 | 4.245 | 0.503 | 6.163 | 0.568 | 5.143 | |
| | 5.000 | 0.100 | 6.223 | 0.330 | 9.151 | 0.376 | 7.785 | |
| | 10.000 | 0.048 | 12.171 | 0.162 | 18.105 | 0.187 | 15.595 | |
| | ∞ | 1.551 | 1.847 | 1.790 | 1.598 | 1.190 | 0.469 | |
| 7 | 1.000 | 0.913 | 1.595 | 2.002 | 1.870 | 2.002 | 1.595 | 0.913 |
| | 0.900 | 0.816 | 1.362 | 2.089 | 1.722 | 2.202 | 1.581 | 1.206 |
| | 0.800 | 0.811 | 1.150 | 2.262 | 1.525 | 2.465 | 1.464 | 1.538 |
| | 0.700 | 0.857 | 0.967 | 2.516 | 1.323 | 2.802 | 1.307 | 1.910 |
| | 0.600 | 0.943 | 0.803 | 2.872 | 1.124 | 3.250 | 1.131 | 2.359 |
| | 0.500 | 1.080 | 0.650 | 3.382 | 0.928 | 3.875 | 0.947 | 2.948 |
| | 0.400 | 1.297 | 0.507 | 4.156 | 0.735 | 4.812 | 0.758 | 3.790 |
| | 0.300 | 1.669 | 0.372 | 5.454 | 0.546 | 6.370 | 0.568 | 5.148 |
| | 0.200 | 2.242 | 0.242 | 8.057 | 0.360 | 9.484 | 0.378 | 7.802 |
| | 0.100 | 4.701 | 0.119 | 15.872 | 0.178 | 18.818 | 0.188 | 15.652 |
| | ∞ | 1.559 | 1.867 | 1.866 | 1.765 | 1.563 | 1.161 | 0.456 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ | $L_5$ | $C_6$ | $L_7$ |
|---|---|---|---|---|---|---|---|---|

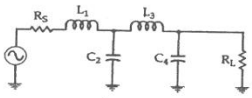


**Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple**

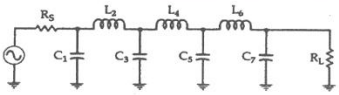



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ |
|---|---|---|---|---|---|
| 2 | 1.355 | 1.209 | 1.638 | | |
| | 1.429 | 0.977 | 1.982 | | |
| | 1.667 | 0.733 | 2.489 | | |
| | 2.000 | 0.560 | 3.054 | | |
| | 2.500 | 0.417 | 3.827 | | |
| | 3.333 | 0.293 | 5.050 | | |
| | 5.000 | 0.184 | 7.426 | | |
| | 10.000 | 0.087 | 14.433 | | |
| | ∞ | 1.391 | 0.819 | | |
| 3 | 1.000 | 1.433 | 1.594 | 1.433 | |
| | 0.900 | 1.426 | 1.494 | 1.622 | |
| | 0.800 | 1.451 | 1.356 | 1.871 | |
| | 0.700 | 1.521 | 1.193 | 2.190 | |
| | 0.600 | 1.648 | 1.017 | 2.603 | |
| | 0.500 | 1.853 | 0.838 | 3.159 | |
| | 0.400 | 2.186 | 0.660 | 3.968 | |
| | 0.300 | 2.763 | 0.486 | 5.279 | |
| | 0.200 | 3.942 | 0.317 | 7.850 | |
| | 0.100 | 7.512 | 0.155 | 15.466 | |
| | ∞ | 1.513 | 1.510 | 0.716 | |
| 4 | 1.355 | 0.992 | 2.148 | 1.585 | 1.341 |
| | 1.429 | 0.779 | 2.348 | 1.429 | 1.700 |
| | 1.667 | 0.576 | 2.730 | 1.185 | 2.243 |
| | 2.000 | 0.440 | 3.227 | 0.967 | 2.856 |
| | 2.500 | 0.329 | 3.961 | 0.760 | 3.698 |
| | 3.333 | 0.233 | 5.178 | 0.560 | 5.030 |
| | 5.000 | 0.148 | 7.607 | 0.367 | 7.614 |
| | 10.000 | 0.070 | 14.887 | 0.180 | 15.230 |
| | ∞ | 1.511 | 1.768 | 1.455 | 0.673 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ |
|---|---|---|---|---|---|

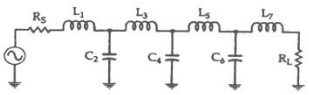



**Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple**

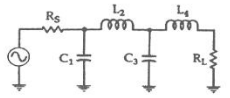



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ | $C_5$ | $L_6$ | $C_7$ |
|---|---|---|---|---|---|---|---|---|
| 5 | 1.000 | 1.301 | 1.556 | 2.241 | 1.556 | 1.301 | | |
| | 0.900 | 1.285 | 1.433 | 2.380 | 1.488 | 1.488 | | |
| | 0.800 | 1.300 | 1.282 | 2.582 | 1.382 | 1.738 | | |
| | 0.700 | 1.358 | 1.117 | 2.868 | 1.244 | 2.062 | | |
| | 0.600 | 1.470 | 0.947 | 3.269 | 1.085 | 2.484 | | |
| | 0.500 | 1.654 | 0.778 | 3.845 | 0.913 | 3.055 | | |
| | 0.400 | 1.954 | 0.612 | 4.720 | 0.733 | 3.886 | | |
| | 0.300 | 2.477 | 0.451 | 6.196 | 0.550 | 5.237 | | |
| | 0.200 | 3.546 | 0.295 | 9.127 | 0.366 | 7.889 | | |
| | 0.100 | 6.787 | 0.115 | 17.957 | 0.182 | 15.745 | | |
| | ∞ | 1.561 | 1.807 | 1.766 | 1.417 | 0.651 | | |
| 6 | 1.355 | 0.942 | 2.080 | 1.659 | 2.247 | 1.534 | 1.277 | |
| | 1.429 | 0.735 | 2.249 | 1.454 | 2.544 | 1.405 | 1.629 | |
| | 1.667 | 0.542 | 2.600 | 1.183 | 3.064 | 1.185 | 2.174 | |
| | 2.000 | 0.414 | 3.068 | 0.958 | 3.712 | 0.979 | 2.794 | |
| | 2.500 | 0.310 | 3.765 | 0.749 | 4.651 | 0.778 | 3.645 | |
| | 3.333 | 0.220 | 4.927 | 0.551 | 6.195 | 0.580 | 4.996 | |
| | 5.000 | 0.139 | 7.250 | 0.361 | 9.261 | 0.384 | 7.618 | |
| | 10.000 | 0.067 | 14.220 | 0.178 | 18.427 | 0.190 | 15.350 | |
| | ∞ | 1.534 | 1.884 | 1.831 | 1.749 | 1.394 | 0.638 | |
| 7 | 1.000 | 1.262 | 1.520 | 2.239 | 1.680 | 2.239 | 1.520 | 1.262 |
| | 0.900 | 1.242 | 1.395 | 2.361 | 1.578 | 2.397 | 1.459 | 1.447 |
| | 0.800 | 1.255 | 1.245 | 2.548 | 1.443 | 2.624 | 1.362 | 1.697 |
| | 0.700 | 1.310 | 1.083 | 2.819 | 1.283 | 2.942 | 1.233 | 2.021 |
| | 0.600 | 1.417 | 0.917 | 3.205 | 1.209 | 3.384 | 1.081 | 2.444 |
| | 0.500 | 1.595 | 0.753 | 3.764 | 0.928 | 4.015 | 0.914 | 3.018 |
| | 0.400 | 1.885 | 0.593 | 4.618 | 0.742 | 4.970 | 0.738 | 3.855 |
| | 0.300 | 2.392 | 0.437 | 6.054 | 0.556 | 6.569 | 0.557 | 5.217 |
| | 0.200 | 3.428 | 0.286 | 8.937 | 0.369 | 9.770 | 0.372 | 7.890 |
| | 0.100 | 6.570 | 0.141 | 17.603 | 0.184 | 19.376 | 0.186 | 15.813 |
| | ∞ | 1.575 | 1.858 | 1.921 | 1.827 | 1.734 | 1.379 | 0.631 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ | $L_5$ | $C_6$ | $L_7$ |
|---|---|---|---|---|---|---|---|---|

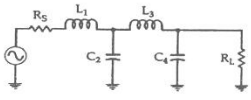



**Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple**

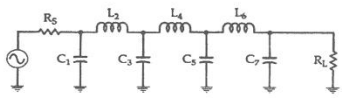



| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ |
|---|---|---|---|---|---|
| 2 | 1.984 | 0.983 | 1.950 | | |
| | 2.000 | 0.909 | 2.103 | | |
| | 2.500 | 0.564 | 3.165 | | |
| | 3.333 | 0.375 | 4.411 | | |
| | 5.000 | 0.228 | 6.700 | | |
| | 10.000 | 0.105 | 13.322 | | |
| | ∞ | 1.307 | 0.975 | | |
| 3 | 1.000 | 1.864 | 1.280 | 1.834 | |
| | 0.900 | 1.918 | 1.209 | 2.026 | |
| | 0.800 | 1.997 | 1.120 | 2.237 | |
| | 0.700 | 2.114 | 1.015 | 2.517 | |
| | 0.500 | 2.557 | 0.759 | 3.436 | |
| | 0.400 | 2.985 | 0.615 | 4.242 | |
| | 0.300 | 3.729 | 0.463 | 5.576 | |
| | 0.200 | 5.254 | 0.309 | 8.225 | |
| | 0.100 | 9.890 | 0.153 | 16.118 | |
| | ∞ | 1.572 | 1.518 | 0.932 | |
| 4 | 1.984 | 0.920 | 2.586 | 1.304 | 1.826 |
| | 2.000 | 0.845 | 2.720 | 1.238 | 1.985 |
| | 2.500 | 0.516 | 3.766 | 0.869 | 3.121 |
| | 3.333 | 0.344 | 5.120 | 0.621 | 4.480 |
| | 5.000 | 0.210 | 7.708 | 0.400 | 6.987 |
| | 10.000 | 0.098 | 15.352 | 0.194 | 14.262 |
| | ∞ | 1.436 | 1.889 | 1.521 | 0.913 |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ |
|---|---|---|---|---|---|




**Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple**




| $n$ | $R_S/R_L$ | $C_1$ | $L_2$ | $C_3$ | $L_4$ | $C_5$ | $L_6$ | $C_7$ |
|---|---|---|---|---|---|---|---|---|
| 5 | 1.000 | 1.807 | 1.303 | 2.691 | 1.303 | 1.807 | | |
| | 0.900 | 1.854 | 1.222 | 2.849 | 1.238 | 1.970 | | |
| | 0.800 | 1.926 | 1.126 | 3.060 | 1.157 | 2.185 | | |
| | 0.700 | 2.035 | 1.015 | 3.353 | 1.058 | 2.470 | | |
| | 0.600 | 2.200 | 0.890 | 3.765 | 0.942 | 2.861 | | |
| | 0.500 | 2.457 | 0.754 | 4.367 | 0.810 | 3.414 | | |
| | 0.400 | 2.870 | 0.609 | 5.296 | 0.664 | 4.245 | | |
| | 0.300 | 3.588 | 0.459 | 6.871 | 0.508 | 5.625 | | |
| | 0.200 | 5.064 | 0.306 | 10.054 | 0.343 | 8.367 | | |
| | 0.100 | 9.556 | 0.153 | 19.647 | 0.173 | 16.574 | | |
| | ∞ | 1.630 | 1.740 | 1.922 | 1.514 | 0.903 | | |
| 6 | 1.984 | 0.905 | 2.577 | 1.368 | 2.713 | 1.299 | 1.796 | |
| | 2.000 | 0.830 | 2.704 | 1.291 | 2.872 | 1.237 | 1.956 | |
| | 2.500 | 0.506 | 3.722 | 0.890 | 4.109 | 0.881 | 3.103 | |
| | 3.333 | 0.337 | 5.055 | 0.632 | 5.699 | 0.635 | 4.481 | |
| | 5.000 | 0.206 | 7.615 | 0.406 | 8.732 | 0.412 | 7.031 | |
| | 10.000 | 0.096 | 15.186 | 0.197 | 17.681 | 0.202 | 14.433 | |
| | ∞ | 1.790 | 1.296 | 2.718 | 1.385 | 2.718 | 1.296 | 1.790 |
| 7 | 1.000 | 1.835 | 1.215 | 2.869 | 1.308 | 2.883 | 1.234 | 1.953 |
| | 0.900 | 1.905 | 1.118 | 3.076 | 1.215 | 3.107 | 1.155 | 2.168 |
| | 0.800 | 2.011 | 1.007 | 3.364 | 1.105 | 3.416 | 1.058 | 2.455 |
| | 0.700 | 2.174 | 0.882 | 3.772 | 0.979 | 3.852 | 0.944 | 2.848 |
| | 0.600 | 2.428 | 0.747 | 4.370 | 0.838 | 2.289 | 0.814 | 3.405 |
| | 0.500 | 2.835 | 0.604 | 5.295 | 0.685 | 5.470 | 0.669 | 4.243 |
| | 0.400 | 3.546 | 0.455 | 6.867 | 0.522 | 7.134 | 0.513 | 5.635 |
| | 0.300 | 5.007 | 0.303 | 10.049 | 0.352 | 10.496 | 0.348 | 8.404 |
| | 0.200 | 9.456 | 0.151 | 19.649 | 0.178 | 20.631 | 0.176 | 16.665 |
| | 0.100 | 1.646 | 1.777 | 2.031 | 1.789 | 1.924 | 1.503 | 0.895 |
| | ∞ | | | | | | | |

| $n$ | $R_L/R_S$ | $L_1$ | $C_2$ | $L_3$ | $C_4$ | $L_5$ | $C_6$ | $L_7$ |
|---|---|---|---|---|---|---|---|---|

